8 Metal peroxides decompose when heated to form metal oxides and oxygen gas. Which factor contributes to solid BaO_2 being more thermally stable than solid MgO_2?

A The hydration enthalpy of Mg^{2+} ion is more exothermic than that of Ba^{2+} ion.

B The lattice energy of BaO_2 is more negative than that of MgO_2.

C The charge density of Ba^{2+} ion is lower than that of Mg^{2+} ion.

D The O–O bond in O_2^{2-} is weaker than the O=O bond in O_2.